

A VCG Mechanism for Transcript Informativeness: Internalizing the Enrollment Externality

Companion to Gans & Kominers (2026), “What Does a Grade Mean?” (NBER WP 35183). Student-side mechanism. Proof-level development.

0. Motivation and the gap being closed

The eigengrade results of Gans–Kominers establish **neutrality**, not **efficiency**, of student course choice. Under exact difficulty adjustment the strategic susceptibility is $\Omega(\mathcal{R}) = 0$, the course-choice bias $b_i(c; \mathcal{R}) = 0$ for every c , and the multinomial-logit choice collapses to the purely pedagogical benchmark $\Pr_i(c; \mathcal{R}) = \Pr_i^*(c)$ (their Props. 7.1, 7.3). Eigengrades delete the *spurious* incentive (arbitrage on $-d_c$) but supply *no* incentive in its place.

Yet the paper’s own precision bound (their Prop. 6.15),

$$\mathbb{E} \|\tilde{\alpha} - (a - \bar{a}\mathbf{1}_n)\|^2 \leq \sigma^2 \operatorname{tr}(L^+) \leq \sigma^2 \frac{n + m - 1}{\lambda_2(L)},$$

shows that every student’s eigengrade variance is a function of the **entire** enrollment graph through the Laplacian spectrum. When student i enrolls in course c she adds edge (i, c) , changes L , and changes the estimation error of **everyone’s** transcript. This is a pure informational externality (a public good): a “bridge” enrollment that strengthens overlap lowers all variances (their Thms. 6.7, 6.15). The neutral mechanism internalizes none of it, so the realized curriculum is pedagogically optimal but informationally suboptimal.

This note builds the mechanism that closes that gap. Eigengrades zero out the wrong incentive; a Groves/VCG transfer then re-introduces the *one correct* incentive — a Pigouvian subsidy equal to the true marginal social informativeness of each enrollment. We prove: dominant-strategy truthfulness (Thm. 2), implementation of the first-best curriculum (Thm. 3), individual rationality under a stated monotone-externality condition (Thm. 4), strict welfare dominance over the neutral outcome (Thm. 5), an exact Pigouvian characterization of the transfer (Prop. 6), and a convex (A-optimal-design) implementation with the honest budget caveat (Prop. 7, Remark 4).

Throughout, transfers are denominated in a currency held **outside** the transcript (enrollment priority points, tuition credits); they must not be written into grades, or they would corrupt the signal the mechanism protects.

1. Setup

1.1 Primitives inherited from the paper

Students $S = \{1, \dots, n\}$, courses $C = \{1, \dots, m\}$, with $N \triangleq n + m$. Latent additive performance when student i takes course c is

$$P_{ic} = a_i - d_c + \varepsilon_{ic}, \quad \mathbb{E}[\varepsilon_{ic}] = 0, \quad \operatorname{Var}(\varepsilon_{ic}) = \sigma^2,$$

with shocks independent across student–course pairs (the noisy-cardinal model underlying their Prop. 6.15). Write $x^* \triangleq (a, d) \in \mathbb{R}^N$, stacking the n student effects above the m course effects.

An **enrollment graph** is $\mathcal{G} = (S \cup C, E)$ with $E \subseteq S \times C$. Each edge $e = (i, c) \in E$ is one observed entry $p_e = P_{ic}$. Let $b_e \in \mathbb{R}^N$ be the **signed incidence vector** of edge $e = (i, c)$: it has $+1$ in student-coordinate i , -1 in course-coordinate $n + c$, and 0 elsewhere, so that

$$b_e^\top x^\star = a_i - d_c.$$

Stacking the rows $b_e^\top$ gives the incidence matrix $B \in \mathbb{R}^{|E| \times N}$ and the bipartite graph Laplacian $L \triangleq B^\top B = \sum_{e \in E} b_e b_e^\top$. The observation model in vector form is $p = Bx^\star + \varepsilon$ with $\text{Cov}(\varepsilon) = \sigma^2 I_{|E|}$.

The universal kernel direction. Let $\mathbf{k} \triangleq \frac{1}{\sqrt{N}}(\mathbf{1}_n, \mathbf{1}_m) \in \mathbb{R}^N$. For *every* edge $e = (i, c)$,

$$b_e^\top \mathbf{k} = \frac{1}{\sqrt{N}} \left(\underbrace{1}_{\text{student } i} - \underbrace{1}_{\text{course } c} \right) = 0,$$

so $\mathbf{k} \in \ker L$ regardless of which edges are present. This is the additive normalization indeterminacy of the paper ($((a, d) \mapsto (a + \kappa \mathbf{1}, d + \kappa \mathbf{1}))$ leaves all $b_e^\top x^\star$ unchanged; their Thm. 6.7). When $\mathcal{G}$ is connected, $\ker L = \text{span}\{\mathbf{k}\}$ exactly, and on $\mathbf{k}^\perp$ the Laplacian is invertible.

1.2 The estimation target and the informational cost

Fix a **target contrast matrix** $K \in \mathbb{R}^{r \times N}$ whose rows lie in $\mathbf{k}^\perp$, so $Kx^\star$ is the object the market wants to learn. Two canonical choices:

- **(T-stud)** $K = M_n$ -rows: the centered student effects $a - \bar{a} \mathbf{1}_n$ (the eigengrade target of their Def. 6.1). Then, on a connected graph, the best linear unbiased estimator (BLUE) of $Kx^\star$ has covariance $\sigma^2 K L^+ K^\top$, and with K the full centered projector this reproduces their $\sigma^2 \text{tr}(L^+)$ bound.
- **(T-course)** $K =$ the centered course effects $d - \bar{d} \mathbf{1}_m$: the *shared* difficulty adjustment that makes every transcript comparable. Its dimension m is fixed independent of n , which is what makes the participation externality of an extra student unambiguously signed (Lemma 1(c) below). We take **(T-course)** as the leading specification and flag where **(T-stud)** differs.

A contrast $k^\top x^\star$ is **estimable** under E iff $k \in \text{row}(B) = \text{col}(L)$. Assume K is estimable under every feasible graph considered (guaranteed when each such graph is connected, since then $\text{col}(L) = \mathbf{k}^\perp \supseteq \text{row}(K)$).

Define the **informational cost** of a graph as the total BLUE variance of the target, in units of σ^2 :

$$\Psi(E) \triangleq \text{tr} \left(\frac{1}{\sigma^2} \text{Cov}(\widehat{Kx^\star}) \right) = \text{tr}(K L(E)^+ K^\top) \quad (\text{connected } E), \quad \Psi(E) \triangleq +\infty \text{ otherwise.}$$

For **(T-stud)** with K the centered projector, $\Psi(E) = \text{tr}(L(E)^+)$ exactly, matching Prop. 6.15.

1.3 The mechanism-design layer

Each student i has a private **pedagogical-value** vector $v_i = (v_{ic})_{c \in C} \in \mathbb{R}_{\geq 0}^m$; v_{ic} is the direct (non-signaling) value of taking course c , exactly the v_{ic} that drives the purely pedagogical benchmark Pr_i^* in their eq. (13). Normalize the outside option (taking no course) to value 0, so $v_{ic} \geq 0$ is without loss after dropping dominated courses. Utility is quasilinear in the transfer:

$$u_i = v_i(E) + t_i, \quad v_i(E) \triangleq \sum_{c: (i,c) \in E} v_{ic}.$$

A **feasibility class** $\mathcal{F}$ collects the admissible enrollment graphs (course capacities, required cores, per-student course loads, and — crucially — connectivity of $\mathcal{G}$, so that Kx^* is estimable and $\Psi < \infty$). Let $\mathcal{F}(S')$ denote the analogous class for the sub-economy with student set $S' \subseteq S$.

The **institution** (“agent 0”) has the publicly known valuation

$$V_0(E) \triangleq -\lambda \sigma^2 \Psi(E), \quad \lambda > 0,$$

where λ is the social price of transcript informativeness (the labor-market value of an accurate $\widehat{Kx^*}$). V_0 contains no private information: it is a known function of E and σ^2 . This is the standard “public component” of a VCG objective and is what lets us run Groves over the students alone.

Social surplus of a graph and its first-best:

$$W(E) \triangleq \sum_{j \in S} v_j(E) + V_0(E) = \sum_{(j,c) \in E} v_{jc} - \lambda \sigma^2 \Psi(E), \quad E^* \in \arg \max_{E \in \mathcal{F}} W(E).$$

The direct mechanism $\mathcal{M}$. Students report $\hat{v} = (\hat{v}_i)_i$. Then:

1. **Decision:** choose $E^*(\hat{v}) \in \arg \max_{E \in \mathcal{F}} \widehat{W}(E)$, where $\widehat{W}(E) = \sum_j \hat{v}_j(E) + V_0(E)$.
2. **Transfer (Groves form):**

$$t_i(\hat{v}) = \underbrace{\sum_{j \neq i} \hat{v}_j(E^*(\hat{v})) + V_0(E^*(\hat{v}))}_{\text{rest-of-society surplus at the chosen graph}} - h_i(\hat{v}_{-i}),$$

with h_i any function of others’ reports. The **Clarke pivot** is

$$h_i(\hat{v}_{-i}) = \max_{E \in \mathcal{F}(S \setminus i)} \left[\sum_{j \neq i} \hat{v}_j(E) + V_0^{-i}(E) \right] = \mathcal{W}(S \setminus i),$$

where $V_0^{-i}(E) = -\lambda \sigma^2 \Psi_{-i}(E)$ uses the Laplacian on vertex set $(S \setminus i) \cup C$, and $\mathcal{W}(S') \triangleq \max_{E \in \mathcal{F}(S')} W_{S'}(E)$ is the optimal welfare of the S' -economy.

2. A linear-model lemma: monotonicity and convexity of Ψ

Everything downstream rests on three facts about Ψ . We prove them through the Gauss–Markov least-norm representation of BLUE variance, which makes the signs transparent.

Lemma 1 (Properties of the informational cost). Consider the linear observation model $p = Bx^* + \varepsilon$, $\text{Cov}(\varepsilon) = \sigma^2 I$, with estimable target Kx^* .

(a) **Least-norm form.** For any estimable contrast $k^\top x^*$, the BLUE variance is

$$\text{Var}(\widehat{k^\top x^*}) = \sigma^2 \nu(k; B), \quad \nu(k; B) \triangleq \min \{\|\ell\|^2 : B^\top \ell = k\}.$$

(b) **Edge-monotonicity.** Adding an observation (edge) weakly lowers Ψ : if B' stacks B with one more row $b_{e'}^\top$, then $\Psi(E \cup \{e'\}) \leq \Psi(E)$. (c) **Participation monotonicity (target T-course).** Adding a student’s edges to the graph weakly lowers Ψ for the fixed course-contrast target. (d) **Convexity in intensities.** Extend to fractional enrollment intensities $z = (z_e)_{e \in \mathcal{E}} \in [0, 1]^\mathcal{E}$ with weighted Laplacian $L(z) = \sum_e z_e b_e b_e^\top$. On the convex set $\mathcal{Z} \triangleq \{z \geq 0 : K \text{ estimable under } z\}$, the map $z \mapsto \Psi(z) = \text{tr}(KL(z)^+ K^\top)$ is convex.

Proof of (a)

A linear estimator $\ell^\top p$ has mean $\mathbb{E}[\ell^\top p] = \ell^\top Bx^* = (B^\top \ell)^\top x^*$. It is unbiased for $k^\top x^*$ at *every* x^* iff $B^\top \ell = k$; the feasible set is nonempty exactly when $k \in \text{col}(B^\top) = \text{row}(B)$, i.e. when $k^\top x^*$ is estimable. Its variance is $\text{Var}(\ell^\top p) = \ell^\top \text{Cov}(p) \ell = \sigma^2 \|\ell\|^2$. The BLUE minimizes variance among unbiased linear estimators, so its variance is $\sigma^2 \min\{\|\ell\|^2 : B^\top \ell = k\} = \sigma^2 \nu(k; B)$. The minimum is attained (a closed convex feasible set, coercive objective), and equals $\sigma^2 k^\top (B^\top B)^+ k = \sigma^2 k^\top L^+ k$ by the normal equations for the least-norm solution; we will not need the closed form. $\square$

Proof of (b)

Stack $B' = \begin{pmatrix} B \\ b_{e'}^\top \end{pmatrix}$ and split any $\ell' \in \mathbb{R}^{|E|+1}$ as $\ell' = (\ell, \ell_{e'})$ with $\ell \in \mathbb{R}^{|E|}$, $\ell_{e'} \in \mathbb{R}$. The constraint $B'^\top \ell' = k$ reads $B^\top \ell + b_{e'} \ell_{e'} = k$. Take any ℓ_0 feasible for the smaller problem, $B^\top \ell_0 = k$. Then $(\ell_0, 0)$ is feasible for the larger problem and has the same norm $\|(\ell_0, 0)\|^2 = \|\ell_0\|^2$. Hence the feasible set of the larger problem contains a norm-preserving image of every point of the smaller problem's feasible set, so its minimum is weakly smaller:

$$\nu(k; B') = \min_{B'^\top \ell' = k} \|\ell'\|^2 \leq \min_{B^\top \ell = k} \|\ell\|^2 = \nu(k; B).$$

Summing over the rows $k = K_{(1)}, \dots, K_{(r)}$ of K gives $\Psi(E \cup \{e'\}) = \sum_s \nu(K_{(s)}; B') \leq \sum_s \nu(K_{(s)}; B) = \Psi(E)$. The inequality is weak because the extra row may be orthogonal to the relevant least-norm direction; it is strict iff some optimal ℓ for the larger problem genuinely uses the new coordinate. $\square$

Proof of (c)

For target **(T-course)**, every row k of K has support only on course-coordinates, so k is the same vector in the S -economy and in the $(S \setminus i)$ -economy (the course block is shared). The $(S \setminus i)$ -economy's incidence matrix B_{-i} is obtained from the S -economy's B by deleting the rows for i 's edges (and the all-zero student-column i , which carries no constraint on a course contrast k since $k_i = 0$). By part (b) applied repeatedly — adding back i 's edges one at a time — Ψ on the larger graph is weakly smaller for each course contrast, hence in the trace. Thus enrolling student i (with any feasible set of edges) weakly lowers the cost of the shared course-contrast target. $\square$

Remark. For target **(T-stud)** part (c) can fail: adding student i introduces a *new* contrast $a_i - \bar{a}$ to be estimated, whose own variance enters Ψ ; the net sign of an extra student on the *total* student-trace is then ambiguous. This is precisely why we adopt the course-contrast public good as the leading specification for the individual-rationality and budget statements below.

Proof of (d)

Fix $z^0, z^1 \in \mathcal{Z}$ and $t \in [0, 1]$; set $z^t = (1-t)z^0 + tz^1$. Because $L(z) = \sum_e z_e b_e b_e^\top$ is linear in z , $L(z^t) = (1-t)L(z^0) + tL(z^1)$. Estimability of K holds on the segment (the column space only grows under nonnegative combination), so write the cost via the information matrix restricted to the estimable subspace. Let Π be the orthogonal projector onto $\mathcal{R} \triangleq \text{row}(K) \subseteq \mathbf{k}^\perp$, and let $A(z) \triangleq \Pi L(z) \Pi$, which is positive definite on $\mathcal{R}$ for $z \in \mathcal{Z}$. On $\mathcal{R}$, $\Psi(z) = \text{tr}(KA(z)^{-1}K^\top)$, and $z \mapsto A(z)$ is affine. It therefore suffices to show $A \mapsto \text{tr}(KA^{-1}K^\top)$ is convex on positive-definite A , since the composition of a convex function with an affine map is convex.

Fix symmetric $A \succ 0$ and symmetric H , and set $A(s) = A + sH \succ 0$ for small s . Differentiating $A(s)A(s)^{-1} = I$ twice:

$$\frac{d}{ds}A(s)^{-1} = -A(s)^{-1}HA(s)^{-1}, \quad \frac{d^2}{ds^2}A(s)^{-1} = 2A(s)^{-1}HA(s)^{-1}HA(s)^{-1}.$$

Evaluate at $s = 0$ and abbreviate $A = A(0)$. Then

$$\left. \frac{d^2}{ds^2} \right|_0 \text{tr}(KA(s)^{-1}K^\top) = 2 \text{tr}(KA^{-1}HA^{-1}HA^{-1}K^\top).$$

Write $A^{-1} = A^{-1/2}A^{-1/2}$ and $S \triangleq A^{-1/2}HA^{-1/2}$ (symmetric). Then $A^{-1}HA^{-1}HA^{-1} = A^{-1/2}S^2A^{-1/2}$, so, using cyclicity of the trace and $S = S^\top$,

$$\text{tr}(KA^{-1/2}S^2A^{-1/2}K^\top) = \text{tr}((SA^{-1/2}K^\top)^\top(SA^{-1/2}K^\top)) = \|SA^{-1/2}K^\top\|_F^2 \geq 0.$$

Hence the second derivative is nonnegative for every direction H , so the map is convex, and Ψ is convex on $\mathcal{Z}$. $\square$

Marginal informativeness (used in Prop. 6). Differentiating $\Psi(z) = \text{tr}(KL(z)^+K^\top)$ in a single intensity z_e (interior, connected support) by the same first-derivative identity gives

$$\frac{\partial \Psi}{\partial z_e} = -\text{tr}(KL^+b_e b_e^\top L^+K^\top) = -\|KL^+b_e\|^2 \leq 0, \quad (*)$$

the (nonpositive) marginal change in target variance from intensifying edge e . For **(T-stud)** with K the centered projector this is $-\|L^+b_e\|^2$, a squared-effective-resistance quantity.

3. Incentive compatibility and efficiency

Theorem 2 (Dominant-strategy truthfulness). In the direct mechanism $\mathcal{M}$ with any Groves pivot $h_i(\hat{v}_{-i})$, truthful reporting $\hat{v}_i = v_i$ is a weakly dominant strategy for every student i , regardless of the reports $\hat{v}_{-i}$ of the others.

Proof

Fix i and arbitrary $\hat{v}_{-i}$. Student i 's realized utility from report $\hat{v}_i$ is

$$u_i(\hat{v}_i) = v_i(E^*) + t_i = v_i(E^*) + \sum_{j \neq i} \hat{v}_j(E^*) + V_0(E^*) - h_i(\hat{v}_{-i}),$$

where $E^* = E^*(\hat{v}_i, \hat{v}_{-i})$ and we use i 's **true** value v_i for her own realized courses (the first term is her actual payoff, not her report). Group the report-independent terms and define

$$\Phi(E) \triangleq \underbrace{v_i(E)}_{\text{true}} + \sum_{j \neq i} \hat{v}_j(E) + V_0(E).$$

Then $u_i(\hat{v}_i) = \Phi(E^*(\hat{v}_i, \hat{v}_{-i})) - h_i(\hat{v}_{-i})$, and h_i does not depend on $\hat{v}_i$. So maximizing u_i over $\hat{v}_i$ is equivalent to choosing the report that induces the graph maximizing Φ .

The mechanism's decision rule selects E^* to maximize $\widehat{W}(E) = \hat{v}_i(E) + \sum_{j \neq i} \hat{v}_j(E) + V_0(E)$ over $E \in \mathcal{F}$. If i reports truthfully, $\hat{v}_i = v_i$, then $\widehat{W} \equiv \Phi$, so the induced $E^*(v_i, \hat{v}_{-i})$ is a maximizer of Φ over $\mathcal{F}$:

$$\Phi(E^*(v_i, \hat{v}_{-i})) = \max_{E \in \mathcal{F}} \Phi(E).$$

For any other report $\hat{v}_i$, the induced graph $E^*(\hat{v}_i, \hat{v}_{-i})$ is some element of $\mathcal{F}$, hence $\Phi(E^*(\hat{v}_i, \hat{v}_{-i})) \leq \max_{E \in \mathcal{F}} \Phi(E)$. Therefore

$$u_i(\hat{v}_i) = \Phi(E^*(\hat{v}_i, \hat{v}_{-i})) - h_i \leq \max_{E \in \mathcal{F}} \Phi(E) - h_i = u_i(v_i),$$

so truthful reporting is a best response to every $\hat{v}_{-i}$, i.e. weakly dominant. $\square$

Intuition. The Groves transfer hands student i the entire rest-of-society surplus (others' pedagogical value **plus** the institution's informational value V_0). Her private objective then coincides, term by term, with total surplus, so she cannot do better than to report the data that let the planner maximize it. The informational public good enters her incentives *only* through V_0 inside the transfer — never through a distortion of her grade.

Theorem 3 (Efficiency / first-best curriculum). In the truthful dominant-strategy equilibrium, the realized graph is a first-best curriculum, $E^*(v) \in \arg \max_{E \in \mathcal{F}} W(E)$. In particular it internalizes the enrollment externality: student i is enrolled in course c on the margin iff her pedagogical value weakly exceeds the marginal social informational cost she would impose elsewhere.

Proof

By Theorem 2 every student reports truthfully, so $\hat{v} = v$ and the decision rule returns $E^*(v) \in \arg \max_{E \in \mathcal{F}} \widehat{W}(E) = \arg \max_{E \in \mathcal{F}} W(E)$, which is the definition of a first best.

For the marginal statement, compare two feasible graphs that differ only in whether edge (i, c) is present, E and $E \cup \{(i, c)\}$. Optimality of E^* implies $(i, c) \in E^*$ only if it does not lower W :

$$W(E^*) - W(E^* \setminus \{(i, c)\}) = v_{ic} - \lambda \sigma^2 [\Psi(E^*) - \Psi(E^* \setminus \{(i, c)\})] \geq 0.$$

By Lemma 1(b) the bracket is ≤ 0 , so $-\lambda \sigma^2 [\dots] \geq 0$ is the (nonnegative) marginal informational *benefit* of the edge. The edge is included exactly when

$$v_{ic} \geq \lambda \sigma^2 [\Psi(E^* \setminus \{(i, c)\}) - \Psi(E^*)] \quad (\leq 0),$$

i.e. always weakly worth it when it also helps measurement, and worth it against a pedagogical sacrifice precisely up to the marginal informativeness it creates. $\square$

Contrast with neutrality. Under the neutral eigengrade rule (their Prop. 7.1) student i enrolls to maximize v_{ic} alone; the informational term is absent from her problem. Theorem 3 adds back exactly the missing term, and only that term.

4. Individual rationality, the Pigouvian transfer, and budget

Theorem 4 (Individual rationality under the Clarke pivot). Adopt target (**T-course**) and the Clarke pivot $h_i = \mathcal{W}(S \setminus i)$. Suppose that for each i , the $(S \setminus i)$ -economy

admits a feasible connected graph (so h_i is finite) and that any such graph remains feasible in the S -economy after adding at least one feasible edge for i (free entry of one student). Then truthful participation yields $u_i \geq 0$ for every student.

Proof

At the truthful profile, $\Phi \equiv W$ in the proof of Theorem 2, so

$$u_i = \max_{E \in \mathcal{F}} W(E) - h_i = \mathcal{W}(S) - \mathcal{W}(S \setminus i).$$

Thus i 's utility equals her **social marginal contribution**. We show $\mathcal{W}(S) \geq \mathcal{W}(S \setminus i)$.

Let $E_{-i}^* \in \arg \max_{E \in \mathcal{F}(S \setminus i)} W_{S \setminus i}(E)$ attain $\mathcal{W}(S \setminus i)$. By hypothesis there is a feasible S -economy graph E' obtained from E_{-i}^* by adding one (or more) feasible edges for student i ; if no edge for i is beneficial, take $E' = E_{-i}^*$ with i unenrolled, which is feasible and connected because the course block and all other students are unchanged. Compare welfares:

$$\begin{aligned} W_S(E') &= \underbrace{\sum_{j \neq i} v_j(E')}_{=\sum_{j \neq i} v_j(E_{-i}^*)} + \underbrace{v_i(E')}_{\geq 0} - \lambda \sigma^2 \Psi_S(E'). \\ &= \sum_{j \neq i} v_j(E_{-i}^*) + v_i(E') - \lambda \sigma^2 \Psi_S(E'). \end{aligned}$$

The pedagogical sum over $j \neq i$ is unchanged because adding i 's edges does not alter which courses the other students take. The term $v_i(E') \geq 0$ since pedagogical values are nonnegative. Finally, by Lemma 1(c) the course-contrast cost cannot increase when i 's edges are added: $\Psi_S(E') \leq \Psi_{S \setminus i}(E_{-i}^*)$. Combining,

$$W_S(E') \geq \sum_{j \neq i} v_j(E_{-i}^*) - \lambda \sigma^2 \Psi_{S \setminus i}(E_{-i}^*) = W_{S \setminus i}(E_{-i}^*) = \mathcal{W}(S \setminus i).$$

Since $E' \in \mathcal{F}(S)$, $\mathcal{W}(S) \geq W_S(E') \geq \mathcal{W}(S \setminus i)$, giving $u_i = \mathcal{W}(S) - \mathcal{W}(S \setminus i) \geq 0$. $\square$

Why the target matters. The proof uses Lemma 1(c), which holds for the fixed-dimensional course target. Under **(T-stud)** the extra student adds her own variance term and the inequality $\Psi_S(E') \leq \Psi_{S \setminus i}(\cdot)$ need not hold; IR then requires the explicit condition that i 's marginal contribution — pedagogical value plus net informational externality minus her own estimation variance — is nonnegative.

Proposition 6 (Pigouvian characterization of the transfer). Hold others' enrollments fixed and consider student i 's marginal incentive to add course c . Her marginal utility under $\mathcal{M}$ is

$$\frac{\partial u_i}{\partial(\text{add}(i, c))} = v_{ic} - \lambda \sigma^2 [\Psi(E) - \Psi(E \setminus \{(i, c)\})],$$

i.e. pedagogical value **plus** a subsidy equal to the true marginal social informativeness of the enrollment. In the intensity relaxation this subsidy is $\lambda \sigma^2 \|KL^+ b_{ic}\|^2 \geq 0$ by (*). The transfer's dependence on the chosen course is exactly the informational externality — never the arbitrage term $-d_c$ that drives the raw-grade bias $b_i(c; \mathcal{R})$ of the paper.

Proof

With others' reports and enrollments fixed, the Groves transfer $t_i = \sum_{j \neq i} v_j(E) + V_0(E) - h_i$ has h_i constant in i 's action and $\sum_{j \neq i} v_j(E)$ constant when only i 's edge (i, c) toggles (it changes

none of the other students' courses). Hence the only term of t_i that responds to adding (i, c) is $V_0(E) = -\lambda\sigma^2\Psi(E)$:

$$\Delta t_i = V_0(E) - V_0(E \setminus \{(i, c)\}) = -\lambda\sigma^2[\Psi(E) - \Psi(E \setminus \{(i, c)\})].$$

Adding $\Delta v_i = v_{ic}$ gives the stated marginal utility. The sign of the subsidy follows from Lemma 1(b): $\Psi(E) - \Psi(E \setminus \{(i, c)\}) \leq 0$, so $-\lambda\sigma^2[\dots] \geq 0$. The intensity form is (*). Because Ψ depends on E only through the Laplacian $L = \sum b_e b_e^\top$, which is built from incidence vectors b_e and **not** from the realized course effects d_c , the course-dependence of t_i carries no $-d_c$ component; it is pure marginal informativeness. $\square$

Reading. The paper moves the world from “ $b_i(c; \mathcal{R}) = -d_c \neq 0$ ” (raw grades: wrong incentive) to “ $b_i(c; \mathcal{R}) = 0$ ” (eigengrades: no incentive). Proposition 6 moves it one step further, to “ $\partial u_i / \partial(i \text{ takes } c) = \text{marginal social value of edge } (i, c)$ ” (VCG: the right incentive). Eigengrade neutrality is the necessary first step — it removes the term that would otherwise contaminate the Pigouvian price — and VCG supplies the second.

Proposition 7 (Welfare dominance over neutrality). Let $E^{\text{ped}} \in \arg \max_{E \in \mathcal{F}} \sum_j v_j(E)$ be a pedagogically-optimal curriculum (the curriculum selected, in expectation, by the neutral eigengrade rule, since there $b_i(c; \mathcal{R}) = 0$ and choice maximizes v). Then $W(E^*) \geq W(E^{\text{ped}})$, with strict inequality whenever $E^{\text{ped}} \notin \arg \max_{E \in \mathcal{F}} W$.

Proof

E^* maximizes W over $\mathcal{F}$ and $E^{\text{ped}} \in \mathcal{F}$, so $W(E^*) \geq W(E^{\text{ped}})$ immediately. If the inequality were an equality, then E^{ped} would attain $\max_{E \in \mathcal{F}} W$, i.e. $E^{\text{ped}} \in \arg \max_E W$; contrapositively, if $E^{\text{ped}} \notin \arg \max_E W$ the inequality is strict. The hypothesis that the pedagogical and informational optima disagree is exactly the case where the enrollment externality bites: there exists a feasible reallocation raising $-\lambda\sigma^2\Psi$ by more than it lowers $\sum_j v_j$. $\square$

Magnitude. The welfare gain is $W(E^*) - W(E^{\text{ped}}) = [\sum_j v_j(E^*) - \sum_j v_j(E^{\text{ped}})] + \lambda\sigma^2[\Psi(E^{\text{ped}}) - \Psi(E^*)] \geq 0$: the first bracket is a (weakly negative) pedagogical sacrifice, the second a (weakly positive, by Lemma 1) informational gain, and optimality of E^* guarantees the gain outweighs the sacrifice.

5. Remarks: implementation, budget, robustness

Remark 1 (Convex / A-optimal implementation). The exact decision $\max_{E \in \mathcal{F}} W(E)$ is combinatorial (selecting a subgraph), but Lemma 1(d) makes its continuous relaxation a convex program: over intensities $z \in \mathcal{Z} \cap [\text{feasible polytope}]$,

$$\max_z \sum_e z_e v_e - \lambda\sigma^2 \text{tr}(KL(z)^+ K^\top)$$

is concave maximization (linear pedagogical term minus convex Ψ), i.e. an **A-optimal experimental-design** problem with a linear value term. It is solvable to global optimality by standard convex methods, and connectivity is enforced automatically because $\Psi(z) \rightarrow \infty$ as the support disconnects (the relevant eigenvalue $\rightarrow 0$). This gives an honest, polynomial-time front end; integral curricula follow by rounding with the usual design guarantees.

Remark 2 (Budget is not balanced — and why). The Clarke pivot delivers DSIC + efficiency + IR (Thms. 2–4) but **not** budget balance. With $h_i = \mathcal{W}(S \setminus i)$,

$$\sum_i t_i = \sum_i \left[\underbrace{\sum_{j \neq i} v_j(E^*) + V_0(E^*) - \mathcal{W}(S \setminus i)}_{\text{others' surplus at } E^*} \right],$$

and because each student confers a *positive* informational externality on the rest (Lemma 1(c)), others' surplus at E^* can exceed their stand-alone optimum $\mathcal{W}(S \setminus i)$, making individual $t_i > 0$: the mechanism may have to **pay** students to internalize the public good, i.e. run a deficit. This is not a defect of the construction but the Green–Laffont (1979)/Hurwicz impossibility surfacing: no mechanism is simultaneously efficient, dominant-strategy, and budget-balanced on a connected type domain for a public-good objective. The deficit is the price of financing transcript informativeness and must be covered from the institution's λ -valued budget. (For purely allocative targets with no participation externality the Clarke surplus is nonpositive and the mechanism is no-deficit; the externality is exactly what breaks that here.)

Remark 3 (Uniqueness of the transfer class). By the Green–Laffont / Holmström characterization, on a connected (smoothly path-connected) type domain the Groves transfers are the *only* transfers implementing the efficient decision in dominant strategies. So conditional on wanting both truthfulness and the first-best curriculum, the informational subsidy of Proposition 6 is forced up to the additive pivot $h_i(\hat{v}_{-i})$; the Clarke choice is the canonical IR-optimizing selection.

Remark 4 (What this does and does not solve). *Solves:* the enrollment-externality inefficiency that survives eigengrade neutrality — students are induced to take informationally valuable (e.g. bridging) courses up to the Pigouvian margin, and truthfully so. *Does not address* (by construction, this being the student side): instructor honesty/precision (the §8 leniency and §9 within-course-compression margins), which require the companion instructor-side Groves mechanism; collusion and false-name (Sybil) enrollment, the classic VCG vulnerabilities, here salient because false enrollments could be used to harvest informational subsidies — a coalition-proof or false-name-proof refinement is the natural next safeguard; and the identification prerequisites of the paper (connectivity, anchoring of ordinal cutpoints), which the feasibility class $\mathcal{F}$ and the estimability of K presuppose.

6. Summary of results

Result	Statement
Lemma 1	Ψ is edge-monotone, participation-monotone (T-course), and convex in intensities; marginal informativeness = $\ KL^+b_e\ ^2$.
Theorem 2	Truthful reporting of pedagogical values is weakly dominant.
Theorem 3	The mechanism implements the first-best curriculum, internalizing the enrollment externality.
Theorem 4	Clarke pivot gives $u_i \geq 0$ (IR) for the course-contrast public good.

Result	Statement
Prop. 6	The transfer is exactly Pigouvian: marginal subsidy = marginal social informativeness, with no $-d_c$ arbitrage term.
Prop. 7	Strict welfare dominance over the neutral eigengrade outcome whenever pedagogy and informativeness disagree.
Remarks	Convex A-optimal implementation; unavoidable budget deficit (Green–Laffont); Groves uniqueness; scope limits.

One-line takeaway. Eigengrades set the spurious course-shopping incentive to zero; a Groves transfer priced at the marginal Laplacian informativeness of each enrollment then turns that clean slate into the first-best curriculum — moving the design from *neutrality* to *efficiency* while keeping truthful course choice a dominant strategy.